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A Characterization of Toric Varieties in Characteristic p

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If X is a smooth toric variety over an algebraically closed field of positive characteristic and L is an invertible sheaf on X, it has been shown by Thomsen that the push-forward of L along the Frobenius morphism of X splits into a direct sum of invertible sheaves. We show that this property characterizes smooth projective toric varieties.

1 Introduction

If X is a smooth toric variety over an algebraically closed field of positive characteristic and L is an invertible sheaf on X, then F_*L, the push-forward of L along the Frobenius morphism of X, splits into a direct sum of invertible sheaves [5, 17]. In this article, we show that this property in fact characterizes smooth projective toric varieties.

Theorem 1.1. Let X be a projective connected scheme over an algebraically closed field k of characteristic $p > 0$. Then X is a smooth toric variety if and only if, for every invertible sheaf L on X, F_*L is a direct sum of invertible sheaves. □

Although the "only if" part of this result has been known already, we include a short proof for completeness (Theorem 2.1). The proof of the "if" part (Theorem 3.1) is a bit more involved: we first reduce to the case when the Picard group of X is free abelian and look at the Cox ring R of X. The assumption on X means that R is flat over R^p, so

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R is regular by Kunz's criterion. It follows that R has to be a polynomial ring, so X is a toric variety.

We may expect Theorem 1.1 to hold more generally for X proper. However, our proof uses the characterization of toric varieties in terms of their Cox rings in the form stated in [13], which requires X to be projective. It is worth noting that similar characterizations have been obtained [4, Corollary 4.4] for X satisfying the A_2 condition: every two points have a common affine neighborhood. These theorems, depending on various results on group actions, have been stated only in characteristic zero; therefore we have to be very careful in applying them to actions of tori in positive characteristic.

One easy corollary of the "if" part of Theorem 1.1 is the existence of nonsplit vector bundles on nontoric smooth projective varieties in characteristic $p > 0$.

2 Proof of the "Only If" Part

We provide a slightly shorter and more explicit proof of Thomsen's result [17]. The proof gives a very short computation even for projective spaces (in which case one can use the Horrocks splitting criterion and the projection formula to prove that the direct image of an invertible sheaf is a direct sum of invertible sheaves). The key point of our approach is to consider Frobenius push-forwards of all invertible sheaves at once.

Let X be a smooth toric variety over an algebraically closed field k. By definition, there is a torus T acting on X with an open free orbit, and X is defined by a combinatorial structure called a fan (cf. [9] for a general reference on toric varieties). We denote the number of rays of the fan defining X by r. For any integer $n \geq 1$, let $F_n : X \to X$ be the "toric Frobenius morphism", the unique endomorphism of X extending the nth power map on T (which can also be seen as the quotient by the kernel K_n of the nth power map). If $\mathrm{char} k = p > 0$, X is naturally defined over $\mathbb{F}_p$ and hence the Frobenius twist $X^{(1)}$ of X is naturally identified with X. In this situation, F_p is identified with the relative Frobenius of X. Note that in any case F_n is finite, flat (because X is smooth), and satisfies $F_n^*(L) \simeq L^n$ for every invertible sheaf L on X.

Theorem 2.1 ([17, Theorem 1]). Let X be as above and let $F = F_n$ for some integer $n \geq 1$. Then, for every invertible sheaf L on X, $F_* L$ is a direct sum of invertible sheaves. Furthermore, we have the following explicit description of the direct summands: let $D \in \mathrm{Pic} X$. Then

$$F_*(\mathcal{O}_X(D)) \simeq \bigoplus_{E \in \mathrm{Pic} X} \mathcal{O}_X(E)^{\oplus m(E,D)}, \tag{1}$$

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where the multiplicity $m(E, D)$ equals the number of points in the set $\{0, 1, \ldots, n-1\}^r \subseteq \mathbb{Z}^r \simeq \mathrm{Div}_T X$ ($\mathrm{Div}_T X$ being the group of T-invariant divisors on X) representing the class $D - nE \in \mathrm{Pic} X$ (i.e., the number of T-invariant divisors in $|D - nE|$ with coefficients $< n$). □

Proof. Let us first prove the theorem in the case when X is complete, and reduce to this case afterward.

First of all, we show that the push-forward $F_*(\mathscr{O}_X(D))$ of a T-equivariant invertible sheaf $\mathscr{O}_X(D)$ is a direct sum of invertible sheaves. The kernel K of F on T is a finite diagonalizable group scheme $\mathrm{Spec} k[M/nM]$ (where M is the character lattice of T) acting on $F_*(\mathscr{O}_X(D))$. I claim that the eigensheaves are invertible. As X is covered by T-invariant affine spaces with standard action, the proof is immediately reduced to the case of $T = \mathbb{G}_m^d$ acting in the standard way on the affine space $\mathbb{A}^d = \mathrm{Spec} k[\mathbb{N}^d]$. In such a situation, we can assume that $D = 0$ and that the T-action on $\mathscr{O}_X(D) = \mathscr{O}_X$ is given by a character $\chi \in M = \mathbb{Z}^d$. We investigate the decomposition of $F_*\mathscr{O}_X$ in this case: $\mathscr{O}_X$ corresponds to the $\mathbb{Z}^d$-graded module $k[\mathbb{N}^d](-\chi)$ over $k[\mathbb{N}^d]$, thus $F_*\mathscr{O}_X$ corresponds to the same thing, but over $k[n \cdot \mathbb{N}^d]$, and, for $\lambda \in (\mathbb{Z}/n\mathbb{Z})^d$ (the character group of K), the λ-eigensheaf corresponds to the $k[n \cdot \mathbb{N}^d]$ submodule $k[n \cdot \mathbb{N}^d](-\lambda')$, where $\lambda' = (\lambda'_1, \ldots, \lambda'_d) \in \mathbb{Z}^d$ is the unique representative of λ with $\chi_i \le \lambda'_i < \chi_i + n$ $(i = 1, \ldots, d)$, which is a free $k[n \cdot \mathbb{N}^d]$-module of rank 1. This shows the claim. Since every invertible sheaf on X is equivariant, we now know that, for every invertible sheaf $\mathscr{O}_X(D)$, there exists a decomposition of $F_*(\mathscr{O}_X(D))$ as in (1), and what is left is to compute the multiplicities.

Observe that $m(E, D)$ depends only on $D - nE$: by the projection formula we have $(F_*(\mathscr{O}_X(D))) \otimes \mathscr{O}_X(-E) = F_*(\mathscr{O}_X(D - nE))$, so $m(E, D) = m(0, D - nE)$. Denote $m(0, D)$ simply by $m(D)$ and apply $h^0(-)$ to both sides of (1):

$$h^0(D) = \sum_{E \in \mathrm{Pic} X} m(E, D) \cdot h^0(E) = \sum_{E \in \mathrm{Pic} X} m(D - nE) \cdot h^0(E). \tag{2}$$

We want to use some generating functions, so we fix a basis $D_1, \ldots, D_\rho$ of $\mathrm{Pic} X$ such that the effective cone lies in the positive orthant (this is possible as X is complete, hence the cone in question is strictly convex) and define

$$S(x) = \sum_{a \in \mathbb{Z}^\rho} h^0\left(\sum a_i D_i\right) x^a \quad \text{and} \quad M(x) = \sum_{a \in \mathbb{Z}^\rho} m\left(\sum a_i D_i\right) x^a.$$

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Then (2) just states that

$$S(x) = M(x) \cdot S(x^n)$$

where we write x^n for $(x_1^n, \dots, x_\rho^n)$.

Let us compute the series S. Consider the map $L : \mathbb{Z}^r = \mathrm{Div}_T X \to \mathrm{Pic} X = \mathbb{Z}^\rho$ (the first identification being given by the basis of "ray" divisors, the second by $D_1, \dots, D_\rho$) taking a T-invariant divisor to its class. Because $h^0(D)$ equals the number of effective T-invariant divisors linearly equivalent to D, we get

$$S(x) = \sum_{b \in \mathbb{Z}^r_{\geq 0}} x^{L(b)} = \prod_{i=1}^{r} \frac{1}{1 - x^{L(e_i)}},$$

$e_1, \dots, e_r$ being the standard basis in $\mathbb{Z}^r$. Therefore

$$M(x) = \frac{S(x)}{S(x^n)} = \prod_{i=1}^{r} \frac{1 - x^{nL(e_i)}}{1 - x^{L(e_i)}} = \prod_{i=1}^{r} \left(1 + x^{L(e_i)} + \cdots + x^{(n-1)L(e_i)}\right),$$

hence $m(D)$ is the number of points $p = \sum a_i e_i$ with $0 \leq a_i < n$ and $L(p) = D$.

We turn to the case where X is not necessarily complete. By adding extra cones, we can embed $i : X \to \overline{X}$ into a smooth complete toric variety $\overline{X}$: first we add cones to make the fan (and hence the corresponding toric variety) complete, then we subdivide the added cones to resolve the singularities (cf. [9, Theorem 11.1.9]). Using the diagram

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & \mathbb{Z}^{\overline{r}-r} & = & \mathbb{Z}^{\overline{r}-r} & & \\
 & & & & \downarrow & & \downarrow & & \\
0 & \to & M & \to & \mathrm{Div}_T \overline{X} & \to & \mathrm{Pic} \overline{X} & \to & 0 \\
 & & \| & & \downarrow & & \downarrow & & \\
0 & \to & M & \to & \mathrm{Div}_T X & \to & \mathrm{Pic} X & \to & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}$$

(where M is the character lattice of T) we see that if the theorem holds for $\overline{X}$, then it also holds for X. ∎

2.1 Vanishing of cohomology of invertible sheaves—a result of Borisov and Hua

Using the above description of Frobenius push-forwards of invertible sheaves on toric varieties, one can easily calculate the ranks of cohomology groups of invertible sheaves.

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This was achieved by [6], using a different method. Our idea is as follows: since the Frobenius maps are affine, we have $H^i(X, L) = H^i(X, F_{n*}L)$, and using Theorem 2.1 we can see that, for n large enough, $F_{n*}L$ is a direct sum of invertible sheaves from a fixed *finite* subset of $\mathrm{Pic}X$. Before we state the result, we need some definitions.

For a subset I of the set $\Delta(1)$ of boundary divisors of X (rays of Δ), let $D_I = \bigcup_{i\in I} D_i$ and $L_I = \mathcal{O}_X(-D_I)$ (the ideal sheaf of D_i). Note that here the D_i is the boundary divisor corresponding to the ray i, and bears no connection with the D_i appearing in the proof of Theorem 2.1. Let $K \subseteq \mathrm{Pic}(X)_{\mathbb{Q}}$ be the convex hull of the classes $[L_I]$ for all $I \subseteq \Delta(1)$. We call a subset $I \subseteq \Delta(1)$ *extremal* if $[L_I]$ is a vertex of K. Let

$$B_k = \{I \subseteq \Delta(1) \mid I \text{ is extremal and } H^k(X, L_I) \neq 0\}, \quad k = 0, 1, \ldots, \dim X.$$

For $I \subseteq \Delta(1)$, let C_I be the translated cone

$$C_I = [L_I] + \mathrm{cone}(\{[L_I] - [L_J] \mid J \subseteq \Delta(1)\}) = [L_I] + \mathrm{cone}(\{[L_I] - [L_J] \mid J \subseteq \Delta(1) \text{ is extremal}\}),$$

where, for a subset $S \subseteq \mathrm{Pic}X$, we denote by $\mathrm{cone}(S)$ the set of all linear combinations of elements of S with nonnegative integer coefficients. Finally, let

$$A_k = \{[L] \in \mathrm{Pic}X \mid H^k(X, L) \neq 0\}, \quad k = 0, 1, \ldots, \dim X.$$

Theorem 2.2 ([6, Section 4]). We have $A_k = \bigcup_{I\in B_k} C_I$. □

Proof. Theorem 2.1 states that $\mathcal{O}_X(E)$ is a direct summand of $F_{n*}(\mathcal{O}_X(D))$ only if

$$[E] \in \frac{n-1}{n}K + \frac{1}{n}[D]. \tag{3}$$

In particular, given D, we have that all E in $F_{n*}\mathcal{O}_X(D)$ lie in K for $n \gg 0$. Since F_n is affine, we have $[D] \in A_k$ if and only if $[E] \in A_k$ for some E such that $\mathcal{O}_X(E)$ appears in $F_{n*}(\mathcal{O}_X(D))$.

Recall that if P is a convex polytope and v is a vertex of P, the *star* of v (denoted by $\mathrm{Star}_P(v)$) is the open subset of P consisting of the relative interiors of all faces of P containing v. Note that

$$\mathrm{Star}_P(v) = \bigcup_{n\gg 0} \left(v + \frac{n-1}{n}(P - v) \right).$$

The above equality together with (3) shows that if $I \subseteq \Delta(1)$ is extremal and $J \subseteq \Delta(1)$ is such that $[L_J] \in \mathrm{Star}_K([L_I])$, then L_J is a direct summand of $F_{n*}L_I$ for $n \gg 0$.

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We claim that if $J \subseteq \Delta(1)$ is *not* extremal, then all the cohomology groups of L_J vanish. For this, we choose an extremal $I \subseteq \Delta(1)$ such that $[L_J] \in \mathrm{Star}_K([L_I])$ and $n > 0$ such that L_J is a direct summand of $F_{n*}L_I$. Since L_I is a direct summand of $F_{n*}L_I$ as well, we see that all the cohomology groups of L_J have to be zero.

Fix a $D \in \mathrm{Pic}X$ and choose an $n \gg 0$ such that $(K + \frac{1}{n}D) \cap \mathrm{Pic}X \subseteq K \cap \mathrm{Pic}X$. Then

$$
\begin{aligned}
[D] \in A_k \text{ iff } & H^k(X, \mathscr{O}_X(D)) \neq 0 \\
& \text{iff } H^k(X, L_I) \neq 0 \text{ for some } I \subseteq \Delta(1) \text{ s.t. } [L_I] \in \frac{n-1}{n}K + \frac{1}{n}[D] \\
& \text{iff } H^k(X, L_I) \neq 0 \text{ for some extremal } I \subseteq \Delta(1) \text{ s.t. } [L_I] \in \frac{n-1}{n}K + \frac{1}{n}[D] \\
& \text{iff } H^k(X, L_I) \neq 0 \text{ for some extremal } I \text{ s.t.} [D] \in [L_I] + (n-1)([L_I] - K).
\end{aligned}
$$

To finish the proof, it suffices to note that C_I is the increasing union

$$C_I = \bigcup_{n \geq 1} ([L_I] + (n-1)([L_I] - K)).$$ ■

Remark 2.3. With some additional care, we can recover the dimensions of cohomology groups $h^k(D) := \dim H^k(X, \mathscr{O}_X(D))$. Recall that B_k is the set of all $I \subseteq \Delta(1)$ such that $H^k(X, L_I) \neq 0$. Note that any $I \in B_k$ is extremal and that, for such I, the number $\mu(D, I)$ of copies of L_I in $F_{n*}(\mathscr{O}_X(D))$ is independent of n for $n \gg 0$. We have

$$
\begin{aligned}
h^k(D) &= \sum_{I \in B_k} h^k(L_I) \cdot \mu(D, I) \\
&= \sum_{I \in B_k} h^k(L_I) \cdot \left(\text{coeff. of } x^{D - nL_I} \text{ in } \prod_i \left(1 + x^{L(e_i)} + \cdots + x^{(n-1)L(e_i)}\right) \right).
\end{aligned}
$$

As L_I are the ideal sheaves of the reduced boundary divisors D_I and $h^i(\mathscr{O}_X) = 0$ for $i > 0$, we have $h^i(L_I) = h^{i-1}(\mathscr{O}_{D_I})$ for $i > 1$, $h^1(L_I) = h^0(\mathscr{O}_{D_I}) - 1$ for $I \neq \emptyset$, and $h^0(L_I) = 0$ for $I \neq \emptyset$. The cohomology groups of $\mathscr{O}_{D_I}$ can be computed using an explicit simplicial complex. □

3 Proof of the "If" Part

Let k be an algebraically closed field of characteristic $p > 0$.

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Theorem 3.1. Let X be a projective and connected scheme over k. Suppose that X satisfies the following condition: for every invertible sheaf L on X, the push-forward F_*L of L along the (absolute) Frobenius morphism of X is a direct sum of invertible sheaves. Then X is a smooth toric variety. □

The following two lemmas show that $\mathrm{Pic}X$ is finitely generated and p-torsion free. Recall that a scheme X of characteristic p is called *Frobenius split* if the canonical map $F^\sharp : \mathcal{O}_X \to F_*\mathcal{O}_X$ is a split monomorphism.

Lemma 3.2. Let X be a proper scheme over k. Suppose that $F_*\mathcal{O}_X$ is a direct sum of invertible sheaves. Then X is smooth and Frobenius split. □

Proof. Since $F_*\mathcal{O}_X$ is a direct sum of invertible sheaves, it is locally free, so F is a finite flat morphism. We check that X is reduced: if $f \in \mathcal{O}_X$ and $f^p = 0$, then f acts trivially on the locally free $\mathcal{O}_X$-module $F_*\mathcal{O}_X$, so $f = 0$. Therefore X is regular by Kunz's criterion [15]: *a reduced local ring R of characteristic $p > 0$ is regular if and only if it is flat over R^p*.

To prove the second assertion, first note that the map $F^\sharp : \mathcal{O}_X \to F_*\mathcal{O}_X$ is locally split (because smooth affine varieties are Frobenius split: see e.g., [7, Proposition 1.1.6]). Let $F_*\mathcal{O}_X = \bigoplus_i L_i$ where the L_i are invertible sheaves. We have $\sum_i \dim_k H^0(X, L_i) = \dim_k H^0(X, F_*\mathcal{O}_X) = \dim_k H^0(X, \mathcal{O}_X) = 1$, so we can assume that $H^0(X, L_0) \simeq k$ and $H^0(X, L_i) = 0$ for $i \neq 0$. It follows that $F^\sharp$ factors through the inclusion $L_0 \hookrightarrow F_*\mathcal{O}_X$. The induced map $\mathcal{O}_X \to L_0$ is a locally split homomorphism between invertible sheaves, hence an isomorphism. ■

Lemma 3.3. Let X be as in Theorem 3.1. Then $(\mathrm{Pic}^0 X)_{\mathrm{red}} = 0$ and $\mathrm{Pic}X$ has no p-torsion. □

Proof. Since X is smooth and Frobenius split by Lemma 3.2, by [12, Theorem 8.2] we know that the abelian variety $(\mathrm{Pic}^0 X)_{\mathrm{red}}$ is Frobenius split as well and therefore ordinary [7, Remark 1.3.9(ii)]. Recall that an abelian variety A is called *ordinary* if $A[p] \simeq (\mathbb{Z}/p\mathbb{Z})^{\dim A}$. Therefore the triviality of $(\mathrm{Pic}^0 X)_{\mathrm{red}}$ will follow from the second assertion.

Suppose now that there exists a nontrivial p-torsion invertible sheaf M on X. For invertible sheaves L and L', let $m(L', L)$ denote the multiplicity of L' as a direct summand of F_*L. If $L'^{\oplus m(L', L)}$ is a direct summand of F_*L, then $(L' \otimes M)^{\oplus m(L', L)}$ is a direct summand of $(F_*L) \otimes M = F_*(L \otimes F^*M) = F_*(L \otimes M^p) = F_*L$, which shows that $m(L' \otimes M, L) \geq m(L', L)$. Applying this to $L' \otimes M$ and M^{-1} in place of L' and M shows that in fact $m(L' \otimes M, L) = m(L', L)$. On the other hand, since the Euler characteristic of an invertible

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sheaf depends only on its numerical class (as seen, e.g., using the Hirzebruch–Riemann–Roch formula) and M is numerically trivial, we have $\chi(L') = \chi(L' \otimes M^i)$ $(i = 0, \dots, p-1)$. Combined with the previous observation, this shows that $\chi(F_*L)$ is divisible by p, but $\chi(F_*L) = \chi(L)$ (because F is affine). It follows that the Euler characteristic of every invertible sheaf on X is divisible by p.

We will now show by induction on r that, for any $r \geq 1$, p^r divides the Euler characteristic of every invertible sheaf on X as well. For $r = 1$ the assertion has been shown above. For the induction step, let L be an invertible sheaf on X. Using our observation that for every L', $m(L' \otimes M, L) = m(L', L)$ $(i = 0, \dots, p-1)$, we can break up the decomposition of F_*L into summands of the form

$$L' \oplus (L' \otimes M) \otimes \cdots \otimes (L' \otimes M^{p-1}).$$

It follows that $\chi(L)$ is a sum of terms of the form $\chi(L') + \chi(L' \otimes M) + \cdots + \chi(L' \otimes M^{p-1}) = p \cdot \chi(L')$. By the induction assumption, every $\chi(L')$ is divisible by p^{r-1}, therefore $\chi(L)$ is divisible by p^r, as claimed.

We deduce that the Euler characteristic of every invertible sheaf has to vanish. This contradicts the fact that, for L ample and a large enough m, $H^i(X, L^m) = 0$ for $i > 0$ and $H^0(X, L^m) \neq 0$, hence $\chi(L) > 0$. We conclude that M cannot exist. ∎

Remark 3.4. The last step of the above proof requires X to be projective. Trying to generalize Theorem 3.1 to proper or proper and A_2 varieties would require producing invertible sheaves of nonzero Euler characteristic. We do not know whether every smooth proper variety possesses an invertible sheaf with nonzero Euler characteristic. If X is proper and A_2, X can be embedded in a proper toric variety [18], and one could restrict invertible sheaves from the ambient variety and use Hirzebruch–Riemann–Roch and intersection theory on the ambient variety to prove that the Euler characteristic would be nonzero. This requires showing that if X is a closed irreducible smooth subvariety of a toric variety, then the cohomology cycle class of X (say in ℓ-adic cohomology) is nonzero. We do not know if this holds in general. □

Let X be as in Theorem 3.1. By Lemma 3.3 and the fact that the quotient $\mathrm{Pic}X/\mathrm{Pic}^0X$ is a finitely generated abelian group, we see that $\mathrm{Pic}X \simeq \Lambda' \oplus H$ where Λ' is free of finite rank and H is finite of order prime to p. By the next lemma (which is easy and well known), we can then find a finite étale cover $\pi : Y \to X$ with Galois group H such that the kernel of the induced map $\pi^* : \mathrm{Pic}X \to \mathrm{Pic}Y$ equals H. Let $\Lambda = \pi^*(\mathrm{Pic}X) \subseteq \mathrm{Pic}Y$ (note that $\Lambda \simeq \Lambda'$, so Λ is free of finite rank).

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Lemma 3.5. Let X be a connected proper algebraic variety and H be a finite subgroup of $\mathrm{Pic}X$ of order prime to the characteristic. Then there exists a finite étale map $f: Y \to X$ with Y connected such that the kernel of the induced map $f^*: \mathrm{Pic}X \to \mathrm{Pic}Y$ equals H. □

Proof. Presenting H as a direct sum of cyclic groups, we easily reduce to the case H cyclic, generated by the class of an n-torsion invertible sheaf L. In this case, we construct Y as the standard cyclic cover $Y = \mathrm{Spec}_X A$ where $A = \bigoplus_{i=0}^{n-1} L^i$ with an algebra structure fixed by choosing an isomorphism $L^n \simeq \mathscr{O}_X$. The projection map $f: Y \to X$ is clearly finite étale, so we only need to check that the kernel of the pull-back map induced by f on the Picard groups consists of the L^j. Under the usual correspondence between coherent sheaves on Y and coherent sheaves of A-modules on X, the pull-back f^*M corresponds to $\bigoplus_{i=0}^{n-1} L^i \otimes M$, which is a free A-module for $M \simeq L^j$ for some j. On the other hand, if $\bigoplus_{i=0}^{n-1} L^i \otimes M$ is a free A-module, then

$$\bigoplus_{i=0}^{n-1} L^i \otimes M \simeq A = \bigoplus_{i=0}^{n-1} L^i$$

as $\mathscr{O}_X$-modules. By the Krull–Schmidt theorem [3, Theorem 3], as invertible sheaves are indecomposable $\mathscr{O}_X$-modules, the summands above have to be the same up to permutation, hence $M \simeq L^j$ for some j. ■

Let L be an invertible sheaf on Y whose isomorphism class belongs to Λ. We claim that $F_* L$ is a direct sum of invertible sheaves whose isomorphism classes also belong to Λ. We have $L \simeq \pi^* M$ for some invertible sheaf M on X. By [8, XIV=XV, Section 1, no. 2, Pr. 2(c)], the diagram

$$\begin{array}{ccc} Y & \xrightarrow{F_Y} & Y \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} \\ X & \xrightarrow{F_X} & X \end{array}$$

is a Cartesian diagram of flat morphisms. By flat base change [11, Proposition 9.3], $F_{Y*} L = F_{Y*} \pi^* M = \pi^* F_{X*} M$ is a direct sum of invertible sheaves.

We denote by $R = \bigoplus_{\lambda \in \Lambda} R_\lambda$ the global section ring of (Y, Λ). More precisely, we choose a collection $L_1, \ldots, L_m$ of invertible sheaves on Y whose classes form a basis of

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Λ, and for $\lambda = \sum a_i[L_i] \in \Lambda$, we define $L_\lambda = L_1^{a_1} \otimes \cdots \otimes L_m^{a_m}$, $R_\lambda = H^0(Y, L_\lambda)$ and let

$$R = \bigoplus_{\lambda \in \Lambda} R_\lambda = \bigoplus_{\lambda \in \Lambda} H^0(Y, L_\lambda),$$

which is a graded ring with multiplication given by tensor product of sections.

Lemma 3.6. Under the above assumptions, the following conditions are satisfied:

1. R is a free R^p-module of finite rank,
2. R is a finitely generated k-algebra,
3. R is regular. □

Proof. (1) For $\mu \in \Lambda/p\Lambda$, let μ' be an arbitrary representative of μ in Λ. Then R decomposes as an R^p-module:

$$R = \bigoplus_{\mu \in \Lambda/p\Lambda} M_\mu \quad \text{where } M_\mu = \bigoplus_{\lambda \in \Lambda} H^0(Y, L_{\mu' + p\lambda})$$

(here L_λ is a chosen invertible sheaf representative of $\lambda \in \Lambda \subseteq \mathrm{Pic} Y$). Using the projection formula, the summands can be presented as follows:

$$M_\mu = \bigoplus_{\lambda \in \Lambda} H^0(Y, (F_* L_{\mu'}) \otimes L_\lambda) = \bigoplus_{i=1}^{p^d} \bigoplus_{\lambda \in \Lambda} H^0(Y, L_{\lambda_i + \lambda}) = \bigoplus_{i=1}^{p^d} R^p(\lambda_i),$$

where $d = \dim Y$ (so $p^d = \deg F_Y$) and $F_* L_{\mu'} = \bigoplus L_{\lambda_i}$. Therefore all M_μ are free of finite rank, and hence also R is a free R^p-module of finite rank.

(2) Note the above reasoning exhibits R as a *graded* R^p-module (though not in a canonical way). We can therefore choose a system $f_i \in R_{\lambda_i}$ $(i = 1, \ldots, k)$ of homogeneous generators of R over R^p. I claim that the f_i generate R as a k-algebra. Let $f \in R_\lambda$ be a homogeneous element of R. By assumption, we can write $f = \sum g_i^p f_i$ for some g_i (which can be assumed to be homogeneous). We can again write $g_i = \sum h_{ij}^p g_j$ and so on, and this process cannot go on forever since Λ is not infinitely p-divisible. (Note the example $R = k[t^{1/p^\infty}]$ which is flat over R^p but not Noetherian.)

(3) This follows from (1) by Kunz's criterion [15] applied to the localizations of R, which are Noetherian by (2). ■

Lemma 3.7. Let Λ be a finitely generated, free abelian group and let $R = \bigoplus_{\lambda \in \Lambda} R_\lambda$ be a Λ-graded domain. Suppose that $m := \bigoplus_{\lambda \neq 0} R_\lambda$ is an ideal of R and that $k := R_0 = R/m$

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is a field. If R is regular, it is isomorphic, as a Λ-graded ring, to $k[t_1, \ldots, t_r]$ with the t_i homogeneous. □

Proof. Since R is regular, $r := \dim R$ is finite and the maximal ideal m is generated by r elements $(x_1, \ldots, x_r)$. We can assume that x_i is homogeneous of degree λ_i (pick a homogeneous basis of the r-dimensional vector space m/m^2 and take homogeneous parts of lifts). Consider the map $\psi : k[t_1, \ldots, t_r] \to R$ sending k to $R_0 \simeq k$ and t_i to x_i. It is easily seen to be surjective by induction on the multidegree. We can do this induction because the set of all λ with $R_\lambda \neq 0$ is a finitely generated monoid with no invertible elements, as by assumption R is a domain and m is an ideal. It is also injective since $r = \dim k[t_1, \ldots, t_r] = \dim R + \operatorname{ht} \ker \psi = r + \operatorname{ht} \ker \psi$. ■

Proof of Theorem 3.1. Let $\pi : Y \to X$ be the finite étale cover constructed by applying Lemma 3.5 to the torsion subgroup H of $\operatorname{Pic} X$, with $\Lambda = \pi^* \operatorname{Pic} X \subseteq \operatorname{Pic} Y$. Combining Lemmas 3.6 and 3.7 we see that the section ring of (Y, Λ) is a polynomial ring. Because Λ contains an ample class (take the pull-back of an ample class on X), Y is toric by [13, Theorem 1.5]. Now $\pi : Y \to X$ is étale, Y is rational, so X is separably rationally connected, hence simply connected by [10, Corollaire 3.6], so π is an isomorphism and X is toric. ■

Remark 3.8. Theorem 1.5 from [13] assumes that the characteristic of k is zero. This is however not necessary. Let us briefly outline the proof of this result in arbitrary characteristic.

As noted by Alper [2, Remark 1.1], Luna's étale slice theorem [16] works without change in case of linearly reductive groups; in particular, it works for tori in characteristic p. Therefore the proof of [14, Theorem 5.1] works without changes other than obvious simplifications as well. It follows that if R is a polynomial ring, we can assume that $R = k[x_1, \ldots, x_r]$ with the x_i homogeneous. Then $X = \operatorname{Proj} \bigoplus_{n \geq 0} R_{n\lambda}$, where λ is an ample class, is a toric variety determined by the combinatorial data of the choice of λ and the gradings of x_i.

Here is an alternative argument for the above-quoted "linearization theorem" [14, Theorem 5.1] in the case of a torus action, provided to us by the referee: let a torus T act on an affine space $\mathbb{A}^n$ such that every T-invariant polynomial function is constant. Then $\mathbb{A}^n$ contains a unique closed orbit Ω, since T is linearly reductive. Since Ω is isomorphic to a quotient torus of T, the ring of regular functions of Ω is generated by units $g_1^{\pm 1}, \ldots, g_m^{\pm 1}$ where the g_i are eigenvectors of T. Using again the linear reductivity of T,

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we may extend each g_i to a polynomial function f_i on $\mathbb{A}^n$, which is an eigenvector of T. Then f_i has no zero on some open T-stable neighborhood of Ω, and hence everywhere as Ω is the unique closed orbit. Thus, f_i is constant; it follows that Ω is just a point. Now apply Lemma 3.7 to conclude that T acts linearly on $\mathbb{A}^n$. □

4 End Remarks

Let X be a smooth projective variety over an algebraically closed field k of characteristic $p > 0$. We denote the absolute Frobenius morphism of X by F. We can study the following properties of X:

(A) For any invertible sheaf L on X, F_*L is a direct sum of invertible sheaves.
(B) $F_*\mathcal{O}_X$ is a direct sum of invertible sheaves.
(C) $F_*^s\mathcal{O}_X$ is a direct sum of invertible sheaves for $s \gg 0$.

Our Theorem 1.1 says that property (A) holds if and only if X is a toric variety.

Let us remark that any ordinary abelian variety satisfies (C) (for all s). Indeed, if X is an ordinary abelian variety, then it is Frobenius split, that is, $\mathcal{O}_X$ is a direct summand of $F_*\mathcal{O}_X$ and hence of all $F_*^s\mathcal{O}_X$. If L is a p^s-torsion invertible sheaf on X, then tensoring $\mathcal{O}_X \to F_*^s\mathcal{O}_X \to \mathcal{O}_X$ by L we obtain that L is a direct summand of $(F_*^s\mathcal{O}_X) \otimes L = F_*^s(\mathcal{O}_X \otimes F^{s*}L) = F_*^s(L^{p^s}) = F_*^s\mathcal{O}_X$, that is, $F_*^s\mathcal{O}_X$ contains as a direct summand the direct sum of all p^s-torsion invertible sheaves. But since X is ordinary, it has exactly $p^{s\cdot\dim X} = \operatorname{rank} F_*^s\mathcal{O}_X$ such invertible sheaves, hence

$$F_*^s\mathcal{O}_X \simeq \bigoplus_{L \in \operatorname{Pic} X[p^s]} L.$$

Another interesting fact is that (B) can hold for special nontoric Fano varieties.

Theorem 4.1 ([1, Theorem 2]). For $p > 2$ and X a smooth quadric of even dimension $2n$, $F_*^s\mathcal{O}_X$ is a direct sum of invertible sheaves if and only if $p|n$ and $s = 1$. □

It would be interesting to have a structure theorem for all varieties satisfying (B) or (C)—for example, that in the case (C) the Albanese map $f : X \to A$ makes X into a toric fibration over an ordinary abelian variety (i.e., X is an *equivariant embedding of an ordinary semiabelian variety*).

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